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DESIGN OF OPTIMAL LOW-THRUST LUNAR POLE-SITTER MISSIONS

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Using a thruster similar to Deep Space 1's NSTAR, pole-sitting low-thrust trajectories are discovered in the vicinity of the L_1 and L_2 libration points. The trajectories are computed with a seventh-degree Gauss-Lobatto collocation scheme that automatically positions thrusting and coasting arcs, and aligns the thruster as necessary to satisfy the problem constraints. The trajectories appear to lie on slightly deformed surfaces corresponding to the L_1 and L_2 halo orbit families. A collocation scheme is also developed that first incorporates spiraling out from low-Earth orbit, and finally spiraling down to a stable lunar orbit for continued uncontrolled surveillance of the lunar south pole. Using direct transcription, the pole-sitting coverage time is maximized to 554.18 days, and the minimum elevation angle associated with the optimal trajectory is 13.0°.

INTRODUCTION

Operations at NASA are currently focused on sustaining a human presence on the Moon by the year 2020. There is interest in establishing a ground station at the lunar south pole. The South Pole-Aitken Basin and Shackleton Crater are thought to contain frozen volatiles, perhaps even water ice, which might be useful for human and energy resources in future expeditions. Through its full libration cycle, the lunar south pole is also viewable from the Earth at certain times, and parts are illuminated by the Sun indefinitely. These "peaks of eternal light" may serve as an important power source for long-term future exploration.

Satellite deployment for continuous surveillance, location of potential landing sites, and long-term communications are important components of all these missions. Most studies utilize multi-satellite constellations for complete south pole coverage. For example, Ely[1] constructed a constellation of three satellites in low-altitude, elliptically inclined lunar orbits, with two vehicles always in view of the south pole. Grebow *et al.*[2] demonstrated that constant communications can be accomplished with two spacecraft in many different combinations of Earth-Moon libration point orbits. (See Hamera *et al.*[3] for a comparison of these two approaches.) However, experience with the design of trajectories in a chaotic system, such as the Restricted Three-Body Problem (RTBP), suggests that constant surveillance might be achieved with just one spacecraft in the presence of a small control input. This fact was confirmed by Ozimek *et al.*,[4] who explored the capabilities of solar sails, comparable to NASA's Millennium Space Technology (ST-9) mission, for continuous south pole surveillance. Lunar pole-sitters were also investigated by West.[5] Unfortunately, the solar sail technology to support these trajectories is still in development. Alternatively, long-duration coverage may be accomplished with one spacecraft and low-thrust propulsion. This option remains virtually unexplored. In fact, after extensive investigation, only two previous studies were located

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in the literature, both focusing on the capabilities of low-thrust engines operating as Earth-based pole-sitters.[6,7]

According to NASA, allocation of a payload space for a small 500 kg spacecraft in a launch to the International Space Station (ISS) is being investigated. NASA is interested in the coverage capabilities of this spacecraft (dry mass 50 kg), if equipped with a thruster similar to Deep Space 1's NSTAR (thrust magnitude 150 mN, specific impulse 1,650 s). Originating in ISS orbit, the entire low-thrust mission is characterized by three distinct phases:

1. Earth-centered, spiral out to the vicinity of the Moon
2. Pole-sitting position maintained for as long as possible
3. Moon-centered, spiral down to an elliptically inclined stable orbit

The end-of-life orbit corresponds to a frozen orbit investigated by Ely,[1] and thus would serve thereafter as part of a larger constellation for continued surveillance and lunar operations.

There are many difficulties in constructing trajectories incorporating low-thrust propulsion, particularly trajectories that remain relatively stationary, such as pole-sitters. Solutions are generally not available *a priori*. The trajectories must somehow be restricted to a bounded region below the south pole. Periodic orbits do not exist because the mass of the spacecraft decreases monotonically with time. At best, solutions inside the bounded region will be nearly periodic and as close to stationary as possible. There is little intuition into the behavior these trajectories, including, for example, the positioning of thrust and coast arcs. Additional challenges also exist in obtaining a complete time-history for the thrust direction. Solving this infinite-dimensional problem under the influence of a nonlinear gravity field presents computational difficulties that have been studied by many researchers. (A useful survey is given by Betts.[8])

Traditionally, the unknown thrust magnitude and direction is specified as part of an *indirect* trajectory optimization problem that locally minimizes a performance index, such as burn time. Indirect methods are advantageous because they yield a relatively low-dimensioned problem with an algebraic control law and constraint equations, that, when satisfied, guarantee local optimality. Solving indirect problems requires only an iterative root-solving procedure, such as Newton's method, and accurate initial guesses often produce rapid convergence. However, in general, the radius of convergence for problems solved with indirect methods is small, usually requiring a very accurate initial guess. Producing an initial guess for the costates, which are not typically physically intuitive, can be very complicated (although transformation relationships are sometimes available).[9] Furthermore, the optimality conditions are often cumbersome to derive as the boundary conditions become increasingly complex. Changing the objective function or adding phases often necessitates a nontrivial re-derivation of the entire problem. Finally, path constraints are difficult to enforce. Despite these disadvantages, however, indirect methods are still used extensively,[10,11] and appreciated for their elegance and beauty.

As an alternative, *direct* transcription approaches use collocation[12] and discretize the entire path.[13,14] While discretization only yields an approximation to the exact optimality conditions, in the limit the Karush-Kuhn-Tucker (KKT) conditions are equivalent to the necessary conditions stipulated by the indirect method.[15] With collocation and direct transcription, path constraints along the entire trajectory, such as restricting the motion of the spacecraft to a region below the lunar south pole, are easily enforced. Once the necessary constraint and gradient information is obtained, a variety of numerical methods is available for computing feasible[16] and/or optimal[17] trajectories. For trade studies, direct methods are also easily adapted for changes in the objective function or adding phases of flight. A larger basin of convergence is observed with collocation. In many cases, arbitrary initial conditions

still yield solutions, thus these methods are extremely useful when there is little intuition about the problem. One possible disadvantage of problems solved with collocation and direct transcription is their large dimensionality. However, with the increasing speed of computers and the efficiency of modern (linear algebra) computer algorithms, these methods are now more tractable. Collocation strategies are also implemented in some capacity in software packages such as COLSYS,[18] AUTO,[19] OTIS,[13] and SOCS.[20]

The low-thrust pole-sitter problem, is, perhaps, best solved using collocation.[4] Initially, the trajectory is split into three phases, and each phase is assumed to be independent of the others. The coverage phase, or phase #2, is designed first. Contours of acceleration on a RTBP gravity gradient plot indicate potential stationary locations for the spacecraft. As demonstrated in Ozimek *et al.*[4] and West,[5] optimal coverage of the south pole occurs in the gravity-well near L_2; however, this investigation examines trajectories near L_1 as well. As an initial guess for the collocation algorithm, the spacecraft is first assumed to maintain an exactly stationary or pole-sitting position inside a bounded region below the lunar south. The bounded region is determined by the desired minimum elevation angle and maximum altitude from the lunar south pole. The number of thrust-coast arcs for the trajectory is predetermined, however, the algorithm can remove unnecessary arcs by reducing the time along these arcs to zero. A nearby feasible solution satisfying the differential equations and problem constraints is computed using a minimum-norm, Newton's method. The algorithm automatically determines when it is necessary to thrust and the appropriate direction for alignment of the thruster. The resulting trajectories are nearly periodic, and appear to lie on surfaces corresponding to the L_1 and L_2 southern halo orbit families.[21] However, the surfaces are slightly deformed to satisfy the problem constraints. Using collocation, a feasible solution is then computed incorporating all three phases, including a spiral out to the coverage orbit and finally spiraling down to the stable lunar orbit upon completion of phase #2. Since collocation problems are easily modified for optimization with direct transcription,[13,14] the feasible solution serves as an initial guess for an algorithm that maximizes the time of phase #2 in SNOPT.[17] The results indicate that continuous coverage can be achieved for periods as long as 554.18 days. The minimum elevation angle associated with the optimal solution is 13.0°.

SOLUTION METHOD

Among several options, perhaps the best method to solve the low-thrust pole-sitter problem is an *implicit integration* scheme. Feasible solutions are computed by allowing the states and controls at points along the entire trajectory to enter the problem as unknown variables. Such a process is especially useful when there is very little intuition about the solution space. Knowledge of a control law is not required; the engine is oriented exactly as needed at every instant to satisfy objective constraints. Unlike *explicit integration* subroutines, where the problem sensitivity depends on only the initial state, a larger convergence radius is expected for implicit schemes. This is especially useful for design in chaotic systems, where a slight change in the initial state could induce large variations and unpredictable behavior downstream. Implicit schemes are very fast and extremely robust, allowing for rapid exploration of the design space. They are also readily adapted for direct optimization. There are several interchangeable names in the literature for this direct optimization process, including direct collocation and direct transcription; the latter terminology is employed hereafter, whereas collocation is used to refer to the more general mathematical procedure of implicit integration.

Collocation

The solution for a controlled dynamical system must satisfy the governing ordinary differential equations (ODEs)

$$\dot{\boldsymbol{x}} = \boldsymbol{f}\left(t, \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{\lambda}\right), \tag{1}$$

where t is the time, $\boldsymbol{x}$ is the state vector, $\boldsymbol{u}$ is the control vector, and $\boldsymbol{\lambda}$ is a problem-dependent parameter vector, that might include mass and/or an arbitrary time interval, for example. (Bold indicates vectors.) A particular solution $\boldsymbol{x}(t)$ is infinite-dimensional, since there are infinitely many values of time over the solution interval. Collocation strategies represent the infinite-dimensional solution as a very large finite set of discrete variables. The following collocation scheme is adapted from Ozimek *et al.*[4]

With collocation, the trajectory is composed of n nodes, and $n-1$ total *node segments*, where a segment is the path that connects two neighboring nodes. (See Figure 1.) Using a seventh-degree Gauss-Lobatto quadrature rule,[22] an interpolating polynomial is constructed to ensure that the segment lies on a continuous trajectory. The polynomial is selected such that the points between the nodes minimize the local truncation error. A seventh-degree polynomial is used and the order of accuracy is 12, thereby increasing the allowable step size between nodes compared to more commonly used lower degree methods.

The endpoint states and controls for the ith node segment, $\boldsymbol{x}_i$, $\boldsymbol{u}_i$, $\boldsymbol{x}_{i+1}$, and $\boldsymbol{u}_{i+1}$, are called *node points*. There are also *internal points*, where the states $\boldsymbol{x}_{i,2}$ and $\boldsymbol{x}_{i,3}$ are allowed to vary. In addition to the node points and internal points, there are three *defect points*: $\boldsymbol{x}_{i,1}$, $\boldsymbol{x}_{i,c}$, and $\boldsymbol{x}_{i,4}$. Several options were investigated and tested for specifying the internal and defect point controls, such as spline equations.[13,23] A simple linear interpolation provides a smooth control history and is more computationally efficient than the alternative methods.

Depending on the problem application, the times corresponding to the node points may vary or remain fixed during numerical procedures, but the times associated with the five internal and defect points are always predetermined in accordance with optimizing the local truncation error. In this analysis, the node times are problem-dependent parameters and are, therefore, included in the formation of $\boldsymbol{\lambda}_i$. For the selected node points and internal points to comply with the equations of motion along the ith segment, the three corresponding defects, i.e.

$$\begin{aligned}
\boldsymbol{\Delta}_{i,1}\left(\boldsymbol{x}_i, \boldsymbol{u_i}, \boldsymbol{x}_{i,2}, \boldsymbol{x}_{i,3}, \boldsymbol{\lambda}_i, \boldsymbol{x}_{i+1}, \boldsymbol{u}_{i+1}\right) &= \boldsymbol{0}, \\
\boldsymbol{\Delta}_{i,c}\left(\boldsymbol{x}_i, \boldsymbol{u_i}, \boldsymbol{x}_{i,2}, \boldsymbol{x}_{i,3}, \boldsymbol{\lambda}_i, \boldsymbol{x}_{i+1}, \boldsymbol{u}_{i+1}\right) &= \boldsymbol{0}, \\
\boldsymbol{\Delta}_{i,4}\left(\boldsymbol{x}_i, \boldsymbol{u_i}, \boldsymbol{x}_{i,2}, \boldsymbol{x}_{i,3}, \boldsymbol{\lambda}_i, \boldsymbol{x}_{i+1}, \boldsymbol{u}_{i+1}\right) &= \boldsymbol{0},
\end{aligned} \tag{2}$$

must be satisfied. Equation (2) forces agreement between $\boldsymbol{f}\left(t, \boldsymbol{x}, \boldsymbol{u}, \boldsymbol{\lambda}\right)$ and the time-derivative of the seventh-degree interpolating polynomial. The full expressions for the seventh-degree defect states and the defects in Eq. (2) are long and require several coefficients. (See Herman.[22]) All of the coefficients, however, are constants and only computed once. For numerical considerations, they are stored in a table for speed and efficiency.

Feasible Solutions

The first step for computing solutions is identifying the problem variables. All the variables are included in the complete design variable vector $\boldsymbol{X}$, including the node states and control, internal node states, and any other variables, such as slack variables and time or mass, as stipulated by the problem. Secondly, the problem constraints are written as $\boldsymbol{F}(\boldsymbol{X}) = \boldsymbol{0}$, paying careful attention to

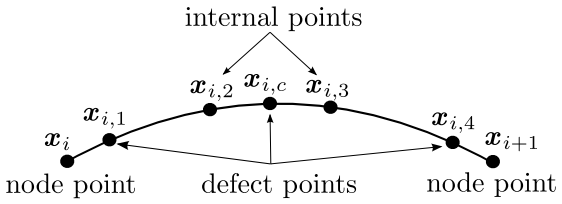


Figure 1. The Seventh-Degree Gauss-Lobatto Node Segment.

each constraint's dependency on $\boldsymbol{X}$. At a minimum, the defect constraints from Eq. (2) must be included in $\boldsymbol{F}$. Path constraints, specific nodal constraints, and other constraints are also included in $\boldsymbol{F}$ as the problem requires; however, there cannot be less variables than constraints. All inequality constraints are converted to equality constraints by introducing slack variables into $\boldsymbol{X}$.

Typically, a solution satisfying $\boldsymbol{F}(\boldsymbol{X}) = \boldsymbol{0}$ is determined by iteratively updating $\boldsymbol{X}$ over j using

$$\boldsymbol{X}^{j+1} = \boldsymbol{X}^j + \alpha \boldsymbol{S}, \tag{3}$$

where α is the scalar step length along the search direction $\boldsymbol{S}$. To determine $\boldsymbol{S}$ and α, consider a Taylor series expansion of $\boldsymbol{F}(\boldsymbol{X}^j) = \boldsymbol{0}$ about $\boldsymbol{X}^j$, and compute $\boldsymbol{X}^{j+1}$ that minimizes $\|\boldsymbol{X}^{j+1} - \boldsymbol{X}^j\|^2$. Then

$$\boldsymbol{S} = -D\boldsymbol{F}\left(\boldsymbol{X}^j\right)^T \left[D\boldsymbol{F}\left(\boldsymbol{X}^j\right) \cdot D\boldsymbol{F}\left(\boldsymbol{X}^j\right)^T\right]^{-1} \boldsymbol{F}\left(\boldsymbol{X}^j\right), \tag{4}$$

and, for $\alpha = 1$, Eq. (3) reduces to a minimum-norm, Newton's method. Given a reasonable initial guess $\boldsymbol{X}^0$, the Newton's method converges quadratically to a nearby solution, provided the solution exists. Once the solution is computed, nodal refinement commonly occurs and the entire process repeats until the error is below a certain tolerance.[24]

In Eq. (4), $D\boldsymbol{F}$ is the Jacobian matrix. In general, $D\boldsymbol{F}$ is very large. However, since $\boldsymbol{\Delta}_{i,1}$, $\boldsymbol{\Delta}_{i,c}$, and $\boldsymbol{\Delta}_{i,4}$ only depend on variables corresponding to the ith node segment, $D\boldsymbol{F}$ is extremely sparse and primarily block diagonal. Thus, most of the entries in $D\boldsymbol{F}$ are zero, and memory is only pre-allocated for the non-zero entries, usually less than 1% of the total size of $D\boldsymbol{F}$. (See Ozimek *et al.*[4] for a detailed discussion on the size and sparsity of $D\boldsymbol{F}$.) Furthermore, there are efficient algorithms available for computing $\left[D\boldsymbol{F} \cdot D\boldsymbol{F}^T\right]^{-1} \boldsymbol{F}$ that exploit the structure of $D\boldsymbol{F}$.[25] All the non-zero elements of $D\boldsymbol{F}$ are computed analyticallly, accept for the derivatives $D\boldsymbol{\Delta}_{i,1}$, $D\boldsymbol{\Delta}_{i,c}$, and $D\boldsymbol{\Delta}_{i,4}$. Since the expressions for $\boldsymbol{\Delta}_{i,1}$, $\boldsymbol{\Delta}_{i,c}$, and $\boldsymbol{\Delta}_{i,4}$ are involved, these derivatives are computed using the complex-step method. The complex-step method is selected for its efficiency and accuracy.[26]

Direct Transcription

After a thorough exploration of the feasible design space, *extremal* solutions are often desired. Obtaining a general extremal trajectory implies the minimization of an objective function of the design variables. This problem can succinctly be posed as

$$\begin{aligned} &\text{Min } J = F_0(\boldsymbol{X}), \\ &\text{subject to } \boldsymbol{F}(\boldsymbol{X}) = \boldsymbol{0}. \end{aligned} \tag{5}$$

The problem is still solved with Eq. (3). However, now α and $\boldsymbol{S}$ must direct $\boldsymbol{X}^{j+1}$ to detect the convex, stationary point associated with the cost function, in addition to satisfying the nonlinear constraints. This type of problem is a nonlinear programming problem (NLP), and there are many

approaches that obtain solutions. In general, however, this parameter optimization formulation does not explicitly involve the Euler-Lagrange constraints and, hence, the objective function is directly minimized without resorting to costate differential equations. It can be demonstrated, however, that the result of the direct method implicitly satisfies the Euler-Lagrange equations.[15]

One NLP method is sequential quadratic programming (SQP). This process first consists of minor iterations for constructing $\boldsymbol{S}$, and then major iterations with Eq. (3). The minor iterations involve solving a quadratic programming (QP) sub-problem. The constraints associated with this QP sub-problems are linearizations of $\boldsymbol{F}(\boldsymbol{X})$, and the objective function of the sub-problem is a quadratic approximation to the Lagrangian function. The sub-problem is posed as

$$\begin{array}{l} \text{Min } Q\left(\boldsymbol{S}^q\right) = F_0(\boldsymbol{X}^j) + DF_0\left(\boldsymbol{X}^j\right)\boldsymbol{S}^q + \frac{1}{2}\boldsymbol{S}^{qT}\boldsymbol{B}\boldsymbol{S}^q, \\ \text{subject to } D\boldsymbol{F}\left(\boldsymbol{X}^j\right)\boldsymbol{S}^q + \beta\boldsymbol{F}\left(\boldsymbol{X}^j\right) = \boldsymbol{0}. \end{array} \tag{6}$$

Here, q is the minor iteration number, $\boldsymbol{B}$ is a positive-definite matrix that approaches the Hessian of the Lagrangian during an iterative procedure, and the scalar β is a problem-dependent parameter. Once $\boldsymbol{S}$ is obtained from the QP sub-problem, the major iterations determine α from a one-dimensional search on the first-order conditions. (For full details on an SQP algorithm, see Gill *et al.*[17])

Almost all of the necessary ingredients for the direct transcription process are available from the formulation of the preceding feasible solution. In fact, the only additional information required is the set of formulas for $F_0(\boldsymbol{X})$ and $DF_0(\boldsymbol{X})$. For most NLP algorithms to solve the direct transcription procedure, including SQP, this first-order information is sufficient. Hence, direct transcription is a natural design transition from the feasible solution method using collocation. As with the feasible solution approach, the efficient handling of the often large, sparse Jacobian matrix $D\boldsymbol{F}$ is crucial. The general-purpose NLP software package SNOPT is one useful tool to solve such problems, while also exploiting the sparse Jacobian matrix structure for economical computation.[17] Also, similar to the feasible solution method, nodal refinement is common with iterations of the entire process to achieve a desired error tolerance.[27]

SYSTEM MODEL

The baseline model for designing nearly pole-sitting trajectories is the Earth-Moon Restricted Three-Body Problem (RTBP), with the addition of low-thrust propulsion. The RTBP is selected because it has proven useful for designing trajectories in the past,[28] and it offers capabilities not available in more simplified models. Of particular interest are the L_1 and L_2 libration points, due to their proximity to the Moon. Control strategies have been developed that require very little control effort, exploiting the chaotic nature of the RTBP to compute trajectories that achieve specific mission design objectives.[29] This suggest that it might be possible to offset trajectories slightly below the L_1 and L_2 points with only a small control input, such as low thrust.

In the Earth-Moon RTBP, it is assumed that the Earth and Moon move in circular orbits, and the spacecraft possesses negligible mass in comparison to the Earth and Moon. A rotating, barycentric coordinate frame is employed, with the x-axis directed from the Earth to the Moon. The z-axis is parallel to the Earth-Moon angular velocity. A low-thrust engine provides additional acceleration $\boldsymbol{a}$ to the system otherwise, for coasting, $\boldsymbol{a} = \boldsymbol{0}$. Then, the equations of motion for the system in the rotating frame are

$$\dot{\boldsymbol{x}} = \boldsymbol{f}\left(t, \boldsymbol{x}, \boldsymbol{u}\right) = \left(\begin{array}{c} \dot{\boldsymbol{r}} \\ \dot{\boldsymbol{v}} \end{array}\right) = \left\{\begin{array}{c} \boldsymbol{v} \\ \boldsymbol{a}\left(t, \boldsymbol{u}\right) - 2\boldsymbol{\Omega} \times \boldsymbol{v} + \boldsymbol{\nabla}^T U\left(\boldsymbol{r}\right) \end{array}\right\}, \tag{7}$$

where the $\boldsymbol{\nabla}^T$ operator refers to gradient-transpose. The components, including position, are derivable from the potential function

$$U = \frac{1-\mu}{\|\boldsymbol{r}-\boldsymbol{r}_1\|} + \frac{\mu}{\|\boldsymbol{r}-\boldsymbol{r}_2\|} + x^2 + y^2, \tag{8}$$

and x, y, and z are the components of the spacecraft's position relative to the rotating, barycentric frame. The mass parameter is μ, the Earth-Moon angular velocity is $\boldsymbol{\Omega}$, and $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ are the positions of the Earth and Moon, respectively. Equation (7) is also nondimensional, where the characteristic quantities are the total mass of the system, the distance between the Earth and Moon, and the magnitude of the system angular velocity.

The magnitude of the thrust acceleration is κ, and it is directed along the vector $\boldsymbol{u}$ with components u_1, u_2, and u_3 relative to the rotating frame. Then

$$\boldsymbol{a} = \kappa\frac{\boldsymbol{u}}{\|\boldsymbol{u}\|} = \frac{T}{m_0 - T\cdot t/c}\frac{\boldsymbol{u}}{\|\boldsymbol{u}\|}, \tag{9}$$

where T is the thrust magnitude, m_0 is the mass of the spacecraft at $t = 0$, and the exhaust velocity is $c = I_{sp}\cdot g_0$. The constant g_0 is the standard gravity acceleration on Earth's surface, and T and I_{sp} are constants determined by the thruster. For engines comparable to Deep Space 1's NSTAR, let $T = 150$ mN and $I_{sp} = 1,650$ s. The constants are nondimensionalized using the appropriate characteristic quantities, including initial spacecraft mass (the dry mass is a characteristic quantity).

APPLICATION TO THE THREE-PHASE POLE-SITTER MISSION SCENARIO

For initial design, the low-thrust mission is separated into three phases. Recall the phases are:

1. Earth-centered, spiral out to the Moon
2. Pole-sitting position maintained for as long as possible
3. Moon-centered, spiral down to an elliptically inclined stable orbit

The phases are also initially assumed to be independent. An algorithm is constructed so that each phase includes a pre-defined number of thrusting arcs, with the addition of path constraints on elevation angle and altitude for phase #2. Since phase #2 is the driving factor for the mission, a feasible solution for phase #2 is first computed with collocation. Once a suitable coverage orbit is determined, the result enters a larger collocation problem that incorporates all three phases. Then an end-of-life feasible solution is computed for the entire mission and optimization maximizes the time to complete phase #2.

Thrusting and Coasting

The basic thrust-coast structure for each phase is depicted in Figure 2, where coast arcs are blue and thrust arcs are red. A similar problem structure appears in Enright,[23] however his algorithm accommodates only two thrust arcs. Here, the user predetermines the number of thrust arcs, k, and a coast arc is always inserted between two thrust arcs. For example, for $k = 2$ there are two thrust arcs separated by one coast arc: the structure is simply *thrust-coast-thrust*. The collocation strategy then shifts the arcs in configuration space as necessary to satisfy the problem constraints, including the optimality conditions for direct transcription.

A relationship between time and initial mass is given by the denominator of Eq. (9). Therefore, the initial mass $m_{0,j}$ for each thrust arc is adjusted accordingly, so that the time is zero at the

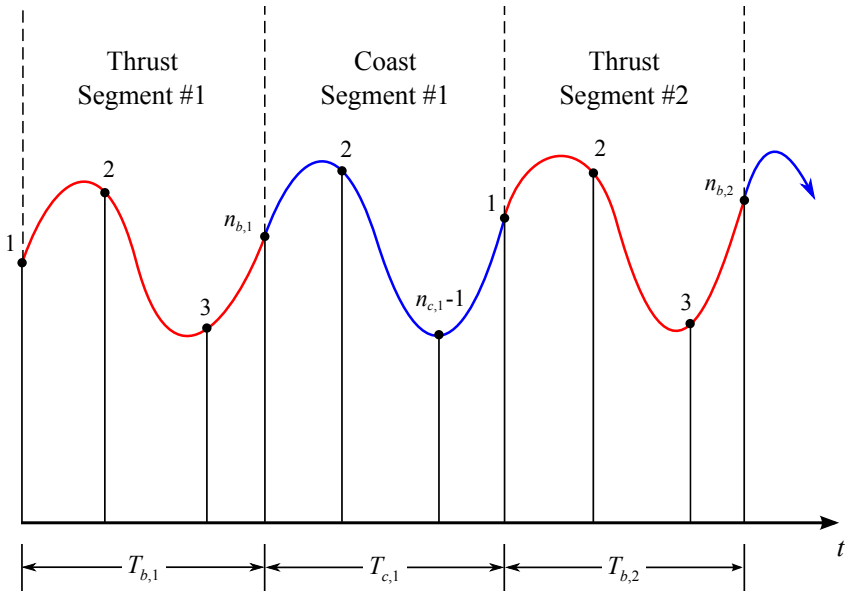


Figure 2. Thrust-Coast Problem Structure.

beginning of the arc. (For coasting, no adjustment is necessary due to time invariance in the RTBP.) The total times along each arc, that is, $T_{b,j}$ and $T_{c,j}$, are specified as problem variables, and so the strategy is capable of removing unnecessary arcs by reducing $T_{b,j}$ or $T_{c,j}$ to zero. Inequality constraints ensure that $T_{b,j}$ and $T_{c,j}$ remain nonnegative. The black dots along the trajectory in Figure 2 represent nodes, with $n_{b,j}$ indicating the number of nodes for the jth thrust arc and similarly for $n_{c,j}$. Each value of $n_{b,j}$ and $n_{b,c}$ is predetermined, so that the number of nodes per arc is a user-defined input. The shared node that connects thrust and coast arcs is formulated as part of the thrust arc. Node times are specified as a fixed ratio of the total time for each arc. For example, for the jth thrust arc, the set of times for each node is $T_{b,j} \times \{0, \delta_2, \delta_3, ..., \delta_{n_{b,j}-2}, \delta_{n_{b,j}-1}, 1\}$, where the ratios δ_i are such that $0 < \delta_2 < \delta_3 < \cdots < \delta_{n_{b,j}-2} < \delta_{n_{b,j}-1} < 1$. For each arc, the time ratios δ_i are fixed, but may be different for different arcs. The number and spacing of the nodes is determined by the accuracy desired for the solution. In general, accuracy is gained by increasing the number of nodes per arc at the expense of computation time.

For thrust arcs, the problem dependent parameters in Eq. (2) are $\boldsymbol{\lambda}_i = (T_{b,j}, m_{0,j})^T$. Thus, $T_{b,j}$ and $m_{0,j}$ are assumed to be *independent* variables for each *node segment*. For coast arcs, the problem dependent parameter is just $\boldsymbol{\lambda}_i = T_{c,j}$. Consequently, constraint equations must be applied to enforce the requirement that $\boldsymbol{\lambda}_i$ be the same for each node segment along the arc. The constraint equations are imposed on adjacent node segments, or

$$\boldsymbol{h}_l = \boldsymbol{\lambda}_l - \boldsymbol{\lambda}_{l-1} = \mathbf{0}. \tag{10}$$

Here, $l = 2, .., n_{b,j} - 1$ and $l = 2, .., n_{c,j} - 1$, for thrusting and coasting, respectively. Formulating the problem in this manner may appear nonintuitive, seeming to increase the size of the problem unnecessarily. However, assuming independent vectors $\boldsymbol{\lambda}_i$ significantly increases the sparsity of $D\boldsymbol{F}$ and $D\boldsymbol{F} \cdot D\boldsymbol{F}^T$. Computing $D\boldsymbol{F}$ is also more tractable, since now all the constraints do not

depend on single variables representing $T_{b,j}$ and $m_{0,j}$, or $T_{c,j}$. Manipulating the dependencies in this way can have a considerable impact on the structure of $D\boldsymbol{F}$, and may mean the difference between a program that requires a few seconds to complete versus one that terminates only after many hours.

Constraints are also imposed on each arc. To ensure that $T_{b,j}$ and $T_{c,j}$ remain nonnegative and mass is continuous, enforce

$$\begin{array}{rrll} c_{b,j} = & -T_{b,j} + \nu_{b,j}^2 & = 0, & \text{for } j = 1, .., k, \\ c_{c,j} = & -T_{c,j} + \nu_{c,j}^2 & = 0, & \text{for } j = 1, .., k-1, \\ \psi_{b,j} = & m_{0,j} - (m_{0,j-1} - T \cdot T_{b,j-1}/c) & = 0, & \text{for } j = 2, .., k, \end{array} \tag{11}$$

where $\nu_{b,j}$ and $\nu_{c,j}$ are new slack variables introduced into the problem. Equation (11) is only applied to the last node segment on each arc, i.e., the variables that appear in Eq. (11) correspond to $\boldsymbol{\lambda}_{n_{b,j}-1}$ and $\boldsymbol{\lambda}_{n_{c,j}-1}$.

For all the phases, the problem variables and constraints are composed of those from each thrust and coast arc. Therefore, the construction of $\boldsymbol{X}$ and $\boldsymbol{F}$ is the same for all three phases. That is,

$$\boldsymbol{X}^T = \left(\boldsymbol{Y}_{b,1}^T, \boldsymbol{Y}_{c,1}^T, \boldsymbol{Y}_{b,2}^T, \boldsymbol{Y}_{c,2}^T, ..., \boldsymbol{Y}_{b,k}^T, \nu_{b,1}, \nu_{c,1}, \nu_{b,2}, \nu_{c,2}, ..., \nu_{b,k}\right), \tag{12}$$

and

$$\boldsymbol{F}\left(\boldsymbol{X}\right)^T = \left(\boldsymbol{G}_{b,1}^T, \boldsymbol{G}_{c,1}^T, \boldsymbol{G}_{b,2}^T, \boldsymbol{G}_{c,2}^T, ..., \boldsymbol{G}_{b,k}^T, c_{b,1}, c_{c,1}, c_{b,2}, c_{c,2}, ..., c_{b,k}, \psi_{b,2}, \psi_{b,3}, ..., \psi_{b,k}\right), \tag{13}$$

where the vectors $\boldsymbol{Y}_{b,j}$ and $\boldsymbol{Y}_{c,j}$ are comprised of the variables for the thrust and coast arcs, respectively. Similarly, the constraint vectors for each arc are $\boldsymbol{G}_{b,j}$ and $\boldsymbol{G}_{c,j}$. That is,

$$\begin{aligned} \boldsymbol{Y}_{b,j}^T = \Big(&\boldsymbol{x}_1^T, \boldsymbol{u}_1^T, \boldsymbol{x}_{1,2}^T, \boldsymbol{x}_{1,3}^T, \boldsymbol{\lambda}_1^T, \boldsymbol{x}_2^T, \boldsymbol{u}_2^T, \boldsymbol{x}_{2,2}^T, \boldsymbol{x}_{2,3}^T, \boldsymbol{\lambda}_2^T, ..., \boldsymbol{x}_{n_{b,j}}^T, \boldsymbol{u}_{n_{b,j}}^T, \\ &\boldsymbol{\eta}_1^T, \boldsymbol{\eta}_{1,2}^T, \boldsymbol{\eta}_{1,3}^T, \boldsymbol{\eta}_2^T, \boldsymbol{\eta}_{2,2}^T, \boldsymbol{\eta}_{2,3}^T, ..., \boldsymbol{\eta}_{n_{b,j}}^T\Big), \\ \boldsymbol{Y}_{c,j}^T = \Big(&\boldsymbol{x}_{1,2}^T, \boldsymbol{x}_{1,3}^T, \boldsymbol{\lambda}_1^T, \boldsymbol{x}_2^T, \boldsymbol{x}_{2,2}^T, \boldsymbol{x}_{2,3}^T, \boldsymbol{\lambda}_2^T, \boldsymbol{x}_3^T, ..., \boldsymbol{x}_{n_{c,j}-1,2}^T, \boldsymbol{x}_{n_{c,j}-1,3}^T, \boldsymbol{\lambda}_{n_{c,j}-1}^T, \\ &\boldsymbol{\eta}_{1,2}^T, \boldsymbol{\eta}_{1,3}^T, \boldsymbol{\eta}_2^T, \boldsymbol{\eta}_{2,2}^T, \boldsymbol{\eta}_{2,3}^T, \boldsymbol{\eta}_3^T, ..., \boldsymbol{\eta}_{n_{c,j}-1,2}^T, \boldsymbol{\eta}_{n_{c,j}-1,3}^T\Big), \\ \boldsymbol{G}_{b,j}^T = \Big(&\boldsymbol{\Delta}_{1,1}^T, \boldsymbol{\Delta}_{1,c}^T, \boldsymbol{\Delta}_{1,4}^T, ..., \boldsymbol{\Delta}_{n_{b,j}-1,1}^T, \boldsymbol{\Delta}_{n_{b,j}-1,c}^T, \boldsymbol{\Delta}_{n_{b,j}-1,4}^T, \boldsymbol{h}_1^T, \boldsymbol{h}_2^T, ..., \boldsymbol{h}_l^T, \\ &\boldsymbol{g}_1^T, \boldsymbol{g}_{1,2}^T, \boldsymbol{g}_{1,3}^T, \boldsymbol{g}_2^T, \boldsymbol{g}_{2,2}^T, \boldsymbol{g}_{2,3}^T, ..., \boldsymbol{g}_n^T\Big), \\ \boldsymbol{G}_{c,j}^T = \Big(&\boldsymbol{\Delta}_{1,1}^T, \boldsymbol{\Delta}_{1,c}^T, \boldsymbol{\Delta}_{1,4}^T, ..., \boldsymbol{\Delta}_{n_{c,j}-1,1}^T, \boldsymbol{\Delta}_{n_{c,j}-1,c}^T, \boldsymbol{\Delta}_{n_{c,j}-1,4}^T, \boldsymbol{h}_1^T, \boldsymbol{h}_2^T, ..., \boldsymbol{h}_l^T, \\ &\boldsymbol{g}_{1,2}^T, \boldsymbol{g}_{1,3}^T, \boldsymbol{g}_2^T, \boldsymbol{g}_{2,2}^T, \boldsymbol{g}_{2,3}^T, \boldsymbol{g}_3^T, ..., \boldsymbol{g}_{n_c-1,2}^T, \boldsymbol{g}_{n_c-1,3}^T\Big). \end{aligned} \tag{14}$$

The vectors $\boldsymbol{g}_i$, $\boldsymbol{g}_{i,2}$, and $\boldsymbol{g}_{i,3}$ represent possible path constraints imposed on the trajectory, and the associated slack variables are $\boldsymbol{\eta}_i$, $\boldsymbol{\eta}_{i,2}$, and $\boldsymbol{\eta}_{i,3}$. Notice that the variables in $\boldsymbol{Y}_{c,j}$ begin and end at variables corresponding to node segment *internal points* since the shared end-point nodes are already included in $\boldsymbol{Y}_{b,j}$. Note also that $\boldsymbol{G}_{b,j}$ depends exclusively on $\boldsymbol{Y}_{b,j}$, whereas $\boldsymbol{G}_{c,j}$ depends on $\boldsymbol{Y}_{c,j}$ and also the shared node states between adjacent thrust arcs.

Application to Phases

Coverage Orbit. For the coverage phase, path constraints are imposed on all the node and internal node states to restrict trajectories to a bounded region below the lunar south pole. Consistent

with Ozimek *et al.*,[4] the constraints are

$$\begin{aligned}
\boldsymbol{g}_i\left(\boldsymbol{r}_i, \boldsymbol{\eta}_i\right) &= \begin{Bmatrix} \sin\phi_{lb} + \dfrac{z_i + R_M}{a_i} \\ a_i - a_{ub} \end{Bmatrix} + \boldsymbol{\eta}_i^2 = \mathbf{0}, \\
\boldsymbol{g}_{i,2}\left(\boldsymbol{r}_{i,2}, \boldsymbol{\eta}_{i,2}\right) &= \begin{Bmatrix} \sin\phi_{lb} + \dfrac{z_{i,2} + R_M}{a_{i,2}} \\ a_{i,2} - a_{ub} \end{Bmatrix} + \boldsymbol{\eta}_{i,2}^2 = \mathbf{0}, \\
\boldsymbol{g}_{i,3}\left(\boldsymbol{r}_{i,3}, \boldsymbol{\eta}_{i,3}\right) &= \begin{Bmatrix} \sin\phi_{lb} + \dfrac{z_{i,3} + R_M}{a_{i,3}} \\ a_{i,3} - a_{ub} \end{Bmatrix} + \boldsymbol{\eta}_{i,3}^2 = \mathbf{0},
\end{aligned} \tag{15}$$

where $a_i = \sqrt{\left(x_i - 1 + \mu\right)^2 + y_i^2 + \left(z_i + R_M\right)^2}$ and R_M is the nondimensional mean radius of the Moon. In Eq. (15), the vector $\boldsymbol{\eta}_i^2$ represents the element-wise square of the slack variable $\boldsymbol{\eta}_i$, and of the same dimension. Enforcing a lower bound ϕ_{lb} on elevation angle and upper bound a_{ub} on altitude, the constraints are applied to both thrusting and coasting arcs for phase #2. The total design variable vector $\boldsymbol{X}$ and constraint vector $\boldsymbol{F}$ are then constructed in accordance with Eqs. (12)-(14). Given an appropriate initial guess $\boldsymbol{X}^0$, Eq. (3) and Eq. (4) can be applied iteratively until $\boldsymbol{F}(\boldsymbol{X}) = \mathbf{0}$, thereby computing a feasible solution close to $\boldsymbol{X}^0$ for phase #2.

Transfer Spirals. Spiraling into and out from the coverage orbit utilizes the same algorithm without enforcing the path constraints in Eq. (15). Instead, the path constraint is

$$\begin{aligned}
\boldsymbol{g}_i\left(\boldsymbol{r}_i, \boldsymbol{\eta}_i\right) &= r_{lb} - r_i + \boldsymbol{\eta}_i^2 = \mathbf{0}, \\
\boldsymbol{g}_{i,2}\left(\boldsymbol{r}_{i,2}, \boldsymbol{\eta}_{i,2}\right) &= r_{lb} - r_{i,2} + \boldsymbol{\eta}_{i,2}^2 = \mathbf{0}, \\
\boldsymbol{g}_{i,3}\left(\boldsymbol{r}_{i,3}, \boldsymbol{\eta}_{i,3}\right) &= r_{lb} - r_{i,3} + \boldsymbol{\eta}_{i,3}^2 = \mathbf{0},
\end{aligned} \tag{16}$$

where $r_i = \sqrt{\left(x_i - 1 + \mu\right)^2 + y_i^2 + z_i^2}$. The lower radial bound r_{lb} from the lunar center is set to a value greater than R_M. Note that $\boldsymbol{g}_i$ and $\boldsymbol{\eta}_i$ are scalar-valued as written in Eq. (16). The variable and constraint formulation is equivalent for both phase #1 and phase #3; the only difference is the initial guess $\boldsymbol{X}^0$ supplied by the user. The complete variable and constraint vectors are then assimilated consistent with Eqs. (12)-(14).

Connecting Phases. Once a feasible solution is determined for phase #2, the result enters a larger collocation problem that also incorporates the variables and constraints for phase #1 and phase #3. Let the vector $\boldsymbol{X}_2$ represent the variables from a solution for phase #2, and $\boldsymbol{F}_2$ comprises the corresponding constraints. Similarly, let $\boldsymbol{X}_1$ and $\boldsymbol{X}_3$ be the variables for phases #1 and #3, with respective constraints $\boldsymbol{F}_1$ and $\boldsymbol{F}_3$. Since the phases are initially assumed to be independent, *boundary conditions* are required to ensure (i) that the states, mass, and/or controls match user-specified values at the beginning of phase #1 and end of phase #3, and (ii) that there exists state, mass, and control continuity between each phase. For example, a constraint on the final state of phase #3, as stipulated by (i), forces the final state to match one given by Ely.[1] Dependency between the phases is established by requirement (ii). These boundary conditions are straightforward to formulate, and comprise the constraint vector $\boldsymbol{F}_{\text{BC}}$. Then, to solve the larger collocation problem, the total design variable vector is

$$\boldsymbol{X}^T = \left(\boldsymbol{X}_1^T, \boldsymbol{X}_2^T, \boldsymbol{X}_3^T\right), \tag{17}$$

and entire constraint vector is

$$\boldsymbol{F}^T = \left(\boldsymbol{F}_1^T, \boldsymbol{F}_2^T, \boldsymbol{F}_3^T, \boldsymbol{F}_{\text{BC}}^T\right). \tag{18}$$

Since the phases are initially assumed to be independent, the Jacobian matrix is block diagonal, composed of the Jacobian submatrices for each phase and the super sparse matrix $D\boldsymbol{F}_{\text{BC}}(\boldsymbol{X})$. (See Figure 3 for the structure of the Jacobian matrix.) Thus, $D\boldsymbol{F}(\boldsymbol{X})$ is easily constructed by inserting the submatrices $D\boldsymbol{F}_1(\boldsymbol{X}_1)$, $D\boldsymbol{F}_2(\boldsymbol{X}_2)$, $D\boldsymbol{F}_3(\boldsymbol{X}_3)$ into the appropriate locations, and computing $D\boldsymbol{F}_{\text{BC}}(\boldsymbol{X})$. The resulting matrix $D\boldsymbol{F}(\boldsymbol{X})$ is extremely sparse, primarily composed of block diagonal submatrices that are also sparse.

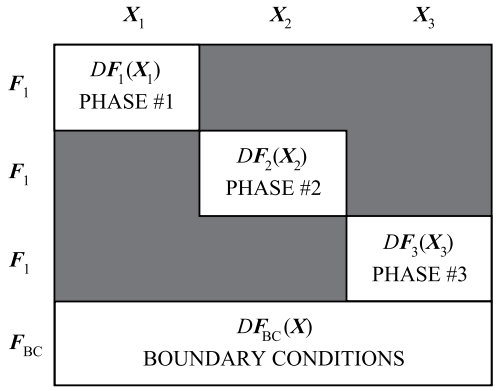


Figure 3. Complete Structure of Jacobian Matrix.

A feasible solution is calculated with Eq. (3) and Eq. (4) using Eq. (17) and Eq. (18). The step length α is adjusted as necessary, however, α is set to one upon entering the basin of attraction corresponding to the numerical method. The feasible solution serves as an initial guess for the optimization with SNOPT, where the time to complete phase #2 is maximized. That is, in Eq. (5), the objective is

$$F_0(\boldsymbol{X}_2) = -\sum_{j=1}^{k} T_{b,j} - \sum_{j=1}^{k-1} T_{c,j}, \tag{19}$$

where $T_{b,j}$ and $T_{c,j}$ are the times for each arc in phase #2, or the coverage phase. The variables that appear in Eq. (19) are the problem dependent variables corresponding to the last node segment along each arc, or $\boldsymbol{\lambda}_{n_{b,j}-1}$ and $\boldsymbol{\lambda}_{n_{c,j}-1}$.

Initial Guess Construction

Coverage Orbit. The ability of the low-thrust spacecraft to achieve suitable line-of-sight coverage in orbit depends on the magnitude of the thrust acceleration κ in Eq. (9). Knowledge of the regions that might contain feasible trajectories for a given value of κ is valuable for predicting the limits of coverage capability, as well as providing an initial guess for a numerical solution. A true pole-sitting trajectory will remain stationary, and the thrust acceleration will exactly offset the gravity gradient experienced by the spacecraft. Trajectories are initially designed under this assumption and later refined to locate nearby feasible solutions, or nearly pole-sitting trajectories. The instantaneous values of $\|\boldsymbol{\nabla} U\|$, as indicated in Figure 4, offer such a means to predetermine the required thrust acceleration and remain stationary in a region given only position information.

As an sample application using Figure 4, consider the preliminary design for the low-thrust lunar pole-sitter coverage orbit, or phase #2. The initial mass upon arrival is usually close to 320 kg, yielding an initial thrust acceleration such that $\kappa = 0.47\,\text{mm}/\text{s}^2$. By Eq. (9), this value is the

lowest thrust acceleration over the duration of the coverage orbit, and thus a conservative estimate of worst-case coverage performance. Inspection of the contours of constant $\|\nabla U\|$ in Figure 4 indicate that a pole-sitting spacecraft is, at least initially, restricted to the red regions that surround the collinear libration points L_1 and L_2. However, recall that since the spacecraft is continuously burning fuel, it cannot remain exactly stationary. In fact, the thrust acceleration increases with time. As indicated by Figure 4, increasing thrust acceleration allows the spacecraft to possibly enter the yellow and green locations that wrap below the lunar south pole. Thus, as phase #2 progresses, the coverage capabilities of the spacecraft increase. Given the proper corrections algorithms to adjust the trajectory, this simple visual inspection approach for estimating the location of potential pole-sitting trajectories is a powerful tool that bypasses the need for more complicated numerical or analytical initial guess schemes.

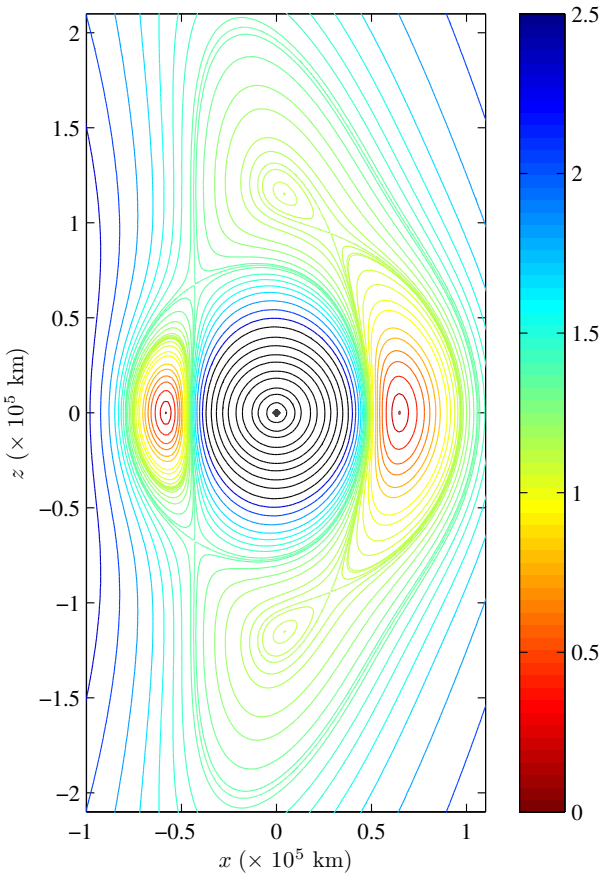


Figure 4. Contours of $\|\nabla U\|$ in mm/s^2, Moon-Centered Rotating Frame.

To formulate the initial guess, the lower bound on elevation angle is fixed at $\phi_{lb} = 13.0^\circ$ and the upper bound on altitude is $a_{ub} = 100,000$ km. For initial design, the spacecraft mass at the beginning of phase #2 is assumed to be 320 kg. This corresponds to $\kappa = 0.47$ mm/s^2. Figure 4 demonstrates that, for this thrust acceleration, the spacecraft is limited to regions near L_1 and L_2. With this in mind, the position variables for the entire trajectory are "stacked" near L_1 or L_2 such that the boundary constraints are satisfied. (Initially, there is no y-component for the trajectory.) Using the stationary assumption, the variables corresponding to velocity are all set to zero. The

thruster is initially aligned strictly in the negative z-direction. The user then specifies the number of thrust arcs k. Since the magnitude of the gravity gradient experienced by the spacecraft remains relatively constant, the nodes are spaced evenly over the arc. The time duration for all the thrust arcs is initially assumed to be equal, and the total thrust time is determined by the user-specified final mass. Here the final mass after completion of the coverage phase is assumed to be 65 kg. The variables corresponding to initial mass are determined by the thrust times. An initial estimate of the total coverage time minus the total thrust time then allows the total coast time to be divided evenly over the number of coast arcs. Now $\boldsymbol{X}_0$ is assimilated in accordance with Eq. (12), where initial guesses for the slack variables are such that the corresponding constraint is initially satisfied.

Alternative approaches to develop the initial guess for the coverage phase are available, however, the previous formulation is perhaps the simplest. The construction best utilizes the pole-sitting assumption, and nearby solutions determined via Eq. (3) and Eq. (4) are not biased by a more complicated non-stationary initial guess. Furthermore, any other initial guess strategy must assume a certain behavior for the solution, and such knowledge is, at this point, unknown.

Transfers. After the feasible solution for phase #2 is computed, the solution enters a larger collocation problem including phases #1 and #3. Generating fully converged transfer trajectories is perhaps the most sensitive of all numerical processes attempted. For this problem, trajectories cannot be determined by simply placing every node at one point in space. Instead, relatively simple numerical integration schemes are utilized to produce the general structure for a trajectory. These schemes are not meant to produce highly accurate guesses, but instead to establish a general time-history and path for the initial nodal distribution. Small refinements may sometimes be available from a simple visual inspection, but further effort is not required. For both transfer phases, i.e., phase #1 and #3, the solution structure is pre-determined as a thrust-coast-thrust sequence ($k = 2$), and a simple two-body inertial velocity-pointing steering law is used during thrusting. This sequence is selected due its emergence in related problems in literature.[30]

The ISS orbit is simulated as a circular orbit in the x-y plane, with an altitude equal to 325 km. Departing this orbit, phase #1 involves hundreds of spirals around the Earth as the spacecraft builds up sufficient energy to escape. To avoid an unnecessary and possibly an intractable number of design variables, including possible problems with poorly scaled variables, an explicit integration process is used exclusively for the majority of the Earth escape. For practical application, the basic velocity-pointing steering law during Earth escape, which maximizes the instantaneous two-body kinetic energy, is operationally simple and is observed to closely track the direction defined by Lawden's primer vector in similar optimal control formulations.[11] This phase #1 spiral sequence terminates once escape from Earth's gravity field is observed, after 202.16 days. (Note that, after this time, a velocity-pointing strategy no longer reflects the fully converged solution as the thrust-direction must support the boundary conditions in phase #2.) At this point, the boundary conditions are stored, and the entire sequence is not considered further as part of the eventual three-phase numerical procedure.

After spiral out from the Earth, the remainder of phase #1 is the final powered Earth-escape leg without spirals, the translunar coast, and the powered insertion into phase #2. These final stages of phase #1 are the only part that is considered for feasibility and optimality in the three-phase solution. (Recall the initial spiral out from Earth is fixed.) There are many ways to create an initial guess for transition into the numerical procedure including a primer-vector law without the optimality constraints or, even more simply, the velocity-pointing law. In this study, a simultaneous forward and backward explicit integration process with the velocity-pointing law is sufficient. A

large discontinuity is observed at the translunar coasting match point, but it is easily resolved in the corrections process.

Construction of the third phase (phase #3) of the trajectory follows a strategy similar to that used to develop phase #1, except that no preliminary spiraling is required; in phase #3 the entire initial guess can be used to initiate the solution process. A variety of explicit integration and visual inspection procedures are available, with a variety of control law predictions. As in the initial guess for phase #1, an inaccurate discontinuity in the path is acceptable. In this case, for lunar orbit capture, the thrusting portion of the thrust-coast-thrust arc employs an anti-velocity pointing law, using the boundary conditions from phase #2 and insertion conditions obtained for a lunar frozen orbit. The frozen orbit that serves as a boundary condition for this study comes from Ely. Insertion occurs at apoapsis of a frozen lunar orbit with a 6,541.4 km semi-major axis and 0.6 eccentricity. The inclination with respect to the x-y plane is 56.2°.[1] Once the integrated guesses for phase #1 and #3 are produced, the paths are decomposed into nodes, and used in conjunction with the coverage orbit solution as an initial guess to construct $\boldsymbol{X}_0$ according to Eq. (17). The collocation scheme detailed previously connects all three phases and so a nearby solution is computed.

NUMERICAL RESULTS

The solution method is successfully applied to the three-phase pole-sitter scenario. The large dimension of the problem demands efficient computational capability, however, visualization is also a key component in the initial guess procedure, and interpretation of the results. As a result, the computing platform is Matlab, but the constraint equations $\boldsymbol{F}$ and Jacobian matrix $D\boldsymbol{F}$ are produced from the MEX-interface with FORTRAN-90 subroutines. The update Eq. (3) occurs in Matlab, where there are extremely efficient algorithms available for computing $\left[D\boldsymbol{F}\cdot D\boldsymbol{F}^T\right]^{-1}\boldsymbol{F}$.[25] All optimization also utilizes a MEX version of SNOPT written as C-subroutines. Phases are first considered independently, with an emphasis on analyzing a variety of candidate solutions for the coverage orbit, i.e., phase #2. When a satisfactory feasible solution for phase #2 is obtained, discontinuous initial guesses for phases #1 and #3 are combined to obtain a fully feasible solution. Finally, this information is used as an initial guess to optimize the coverage duration using the direct transcription process. For reference, the engine parameters are summarized in Table 1.

Table 1. Low-Thrust Engine Parameters.

Parameter	Value	Units
m_0	500	kg
m_f	50	kg
T	150	mN
I_{sp}	1,650	s

Coverage Orbit

By investigating initial guesses near L_1 and L_2, low-thrust trajectories are quickly computed that satisfy a minimum elevation angle of 13.0°. The collocation scheme automatically determines the thruster alignment, and positions the thrusting and coasting arcs as needed. It is observed that the spacecraft thrusts whenever approaching the specified boundary. Hence, thrusting generally occurs at the top of the trajectory and coasting near the bottom. A few candidate results appear in Figure 5, where thrust arcs are red and blue represents a coast arc. There are two characteristic types of solutions. For long duration trajectories, initial estimates of total coverage time are 475 days and 600

days for L_1 and L_2, respectively. These trajectories include a total of 76 thrust arcs and, primarily, the solutions involve an engine pulsing in the negative z-direction, remaining below the libration points. The trajectories appear in the leftmost plots in Figure 5(a). The corrected coverage times are 289.44 days for the L_1 trajectory, and 445.03 days for L_2. Given the large number of thrust arcs, the pulsing solutions require many small thrusting segments, each about 3 days in duration. The solutions might be considered impractical from an operational standpoint, where thrusters most likely require longer, more sustained thrusting and coasting times. Non-pulsing solutions are computed by decreasing the number of total arcs while retaining a total coverage time that is high. For these solutions, the total number of thrust arcs is set to 36, and initial guesses for the coverage times are 425 days and 500 days for L_1 and L_2, respectively. The resulting solutions near L_1 and L_2 can be viewed in the rightmost plots in Figure 5(b). The corrected time for the trajectory near L_1 is 320.22 days, and the L_2 trajectory sustains coverage for 398.56 days. The thrusting segments for these solutions are about 8 days in duration.

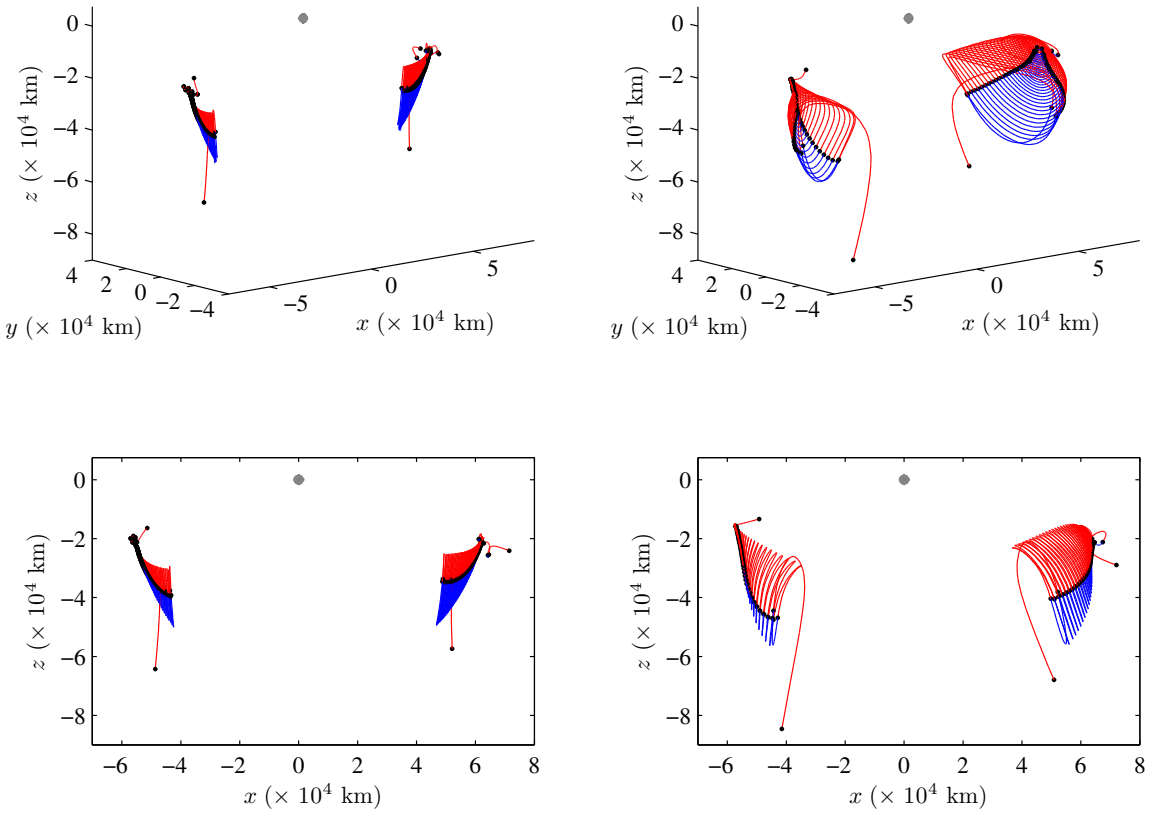


Figure 5. (a) Pulsing and (b) Non-Pulsing Trajectories for the Coverage Phase.

All the solutions move toward the Moon as coverage time increases, as expected due to the increase in thrust acceleration as fuel is consumed. Each solution continuously maintains direct line-of-sight with both the lunar south pole and the Earth. The non-pulsing solutions are similar to the pulsing ones, which appear to expand in the y-direction to satisfy the increase in arc time.

Therefore, whereas motion is primarily in the x-z plane for the pulsing solutions, the non-pulsing solutions appear to be more three-dimensional in nature. A striking feature about the non-pulsing solutions is that the motion seems to be confined to a three-dimensional surface very similar to the surfaces corresponding to the L_1 and L_2 southern halo orbit families.[2,21] The solutions not only appear to move along the family with change in energy, but they also significantly alter the surface shape to allow for the problem constraints.

Since the L_2 non-pulsing solution yields the best results, it is selected as the coverage orbit for the mission design sequence. However, if any of the other trajectories are desired, they can easily be incorporated into the three-phase design without significantly altering the process. The control history for the L_2 non-pulsing solution appears in Figure 6 and the coverage results and thrust acceleration are plotted in Figure 7. The elevation angle results confirm that the spacecraft is always at least 13° above the south pole horizon. As expected, the thrust acceleration increases with time, thereby altering the energy and coverage capabilities of the spacecraft. In fact, the final thrust acceleration is near 1.75 $\mathrm{mm/s^2}$, allowing the spacecraft to enter the light blue regions in Figure 4, and so the trajectory shifts away from the boundary constraint and toward the Moon.

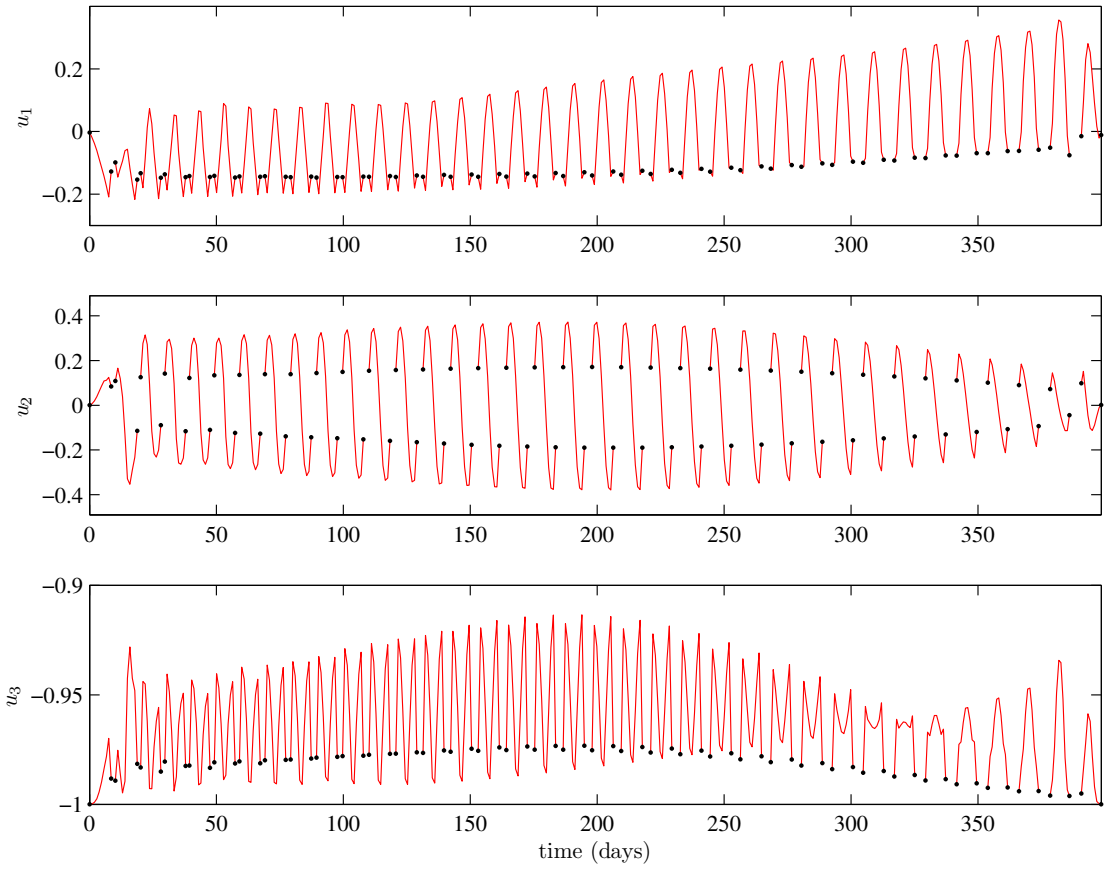


Figure 6. Control History for an L_2 Non-Pulsing Solution.

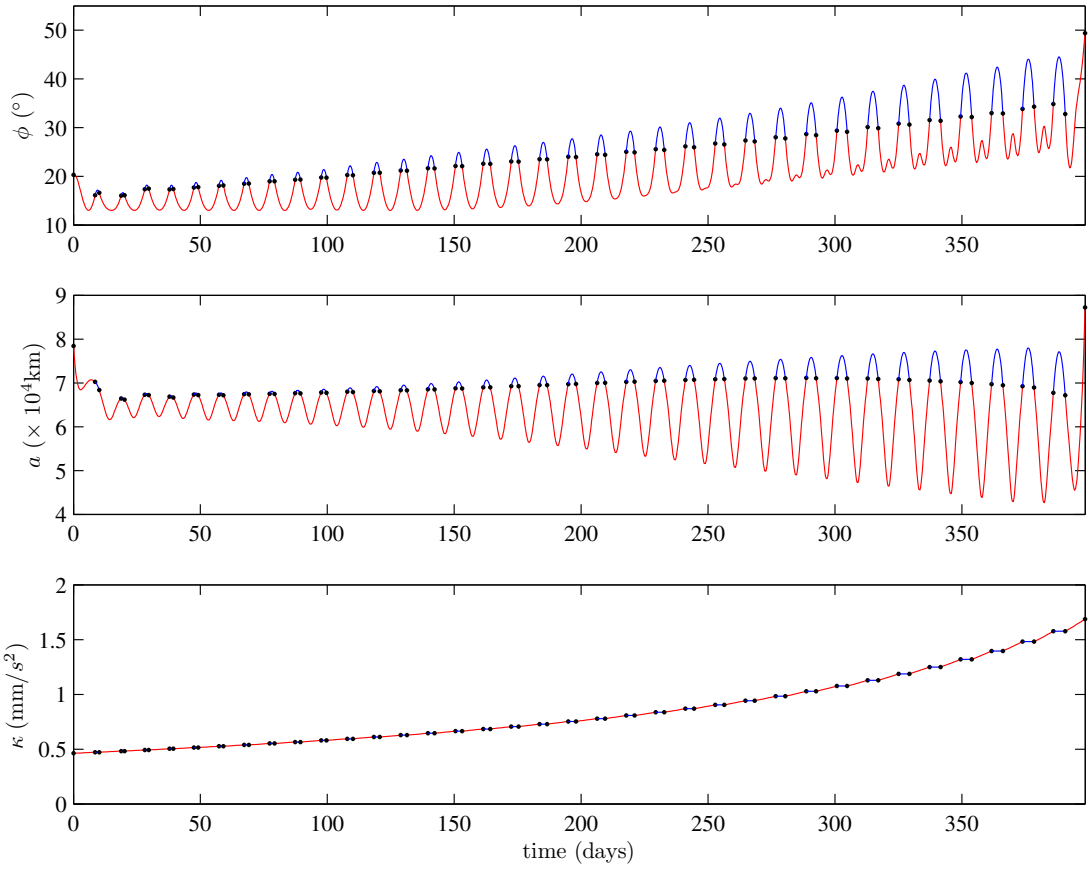


Figure 7. Elevation Angle, Altitude, and Thrust-Acceleration for the L_2 Solution.

Three-Phase Solution

After a thorough exploration of the design space, the solution method is applied simultaneously to all three phases to obtain the desired trajectory. Feasibility is obtained first and used as an initial guess for an optimal solution.

Feasible Solution. A fully feasible solution is obtained by combining all three phases to represent the end-to-end solution. In this process, the continuous, feasible orbit is combined with the discontinuous initial guesses for phase #1 and phase #3. The design variable and constraint vectors from Eq. (17) and Eq. (18), respectively, are used and the feasible solution is iteratively obtained from Eqs. (3)-(4). Even with relatively inaccurate guesses for these discontinuous transfers, the three-phase solution is easily computed, yielding a coverage orbit of 447.04 days. (See Figure 8.) Recall that the spiraling portion denoted in purple is not a part of the corrections process; only the the red thrusting and blue coasting arcs are shifted. The plot also includes a propagation of the final stable lunar orbit (green) upon the completion of phase #3. The total time for the transfer to the Moon, including the fixed transfer-out as well as the time for phase #1, is 244.97 days, corresponding to arrival at the coverage orbit with 315.90 kg of fuel. At this point, the spacecraft utilizes 264.19 kg of fuel to achieve a total of 447.04 days in the phase #2 coverage orbit. Although the coverage orbit uses 35 thrust arcs and 34 coast arcs, the minimum duration for any thrust or coast arc is still 8.85 days. This long arc length implies that the operationally difficult engine pulses are

generally avoided. Since the thrust acceleration is high at the end of phase #2, and the Moon is a smaller body than the Earth, only 2.36 days and 1.76 kg of propellant are necessary to complete phase #3.

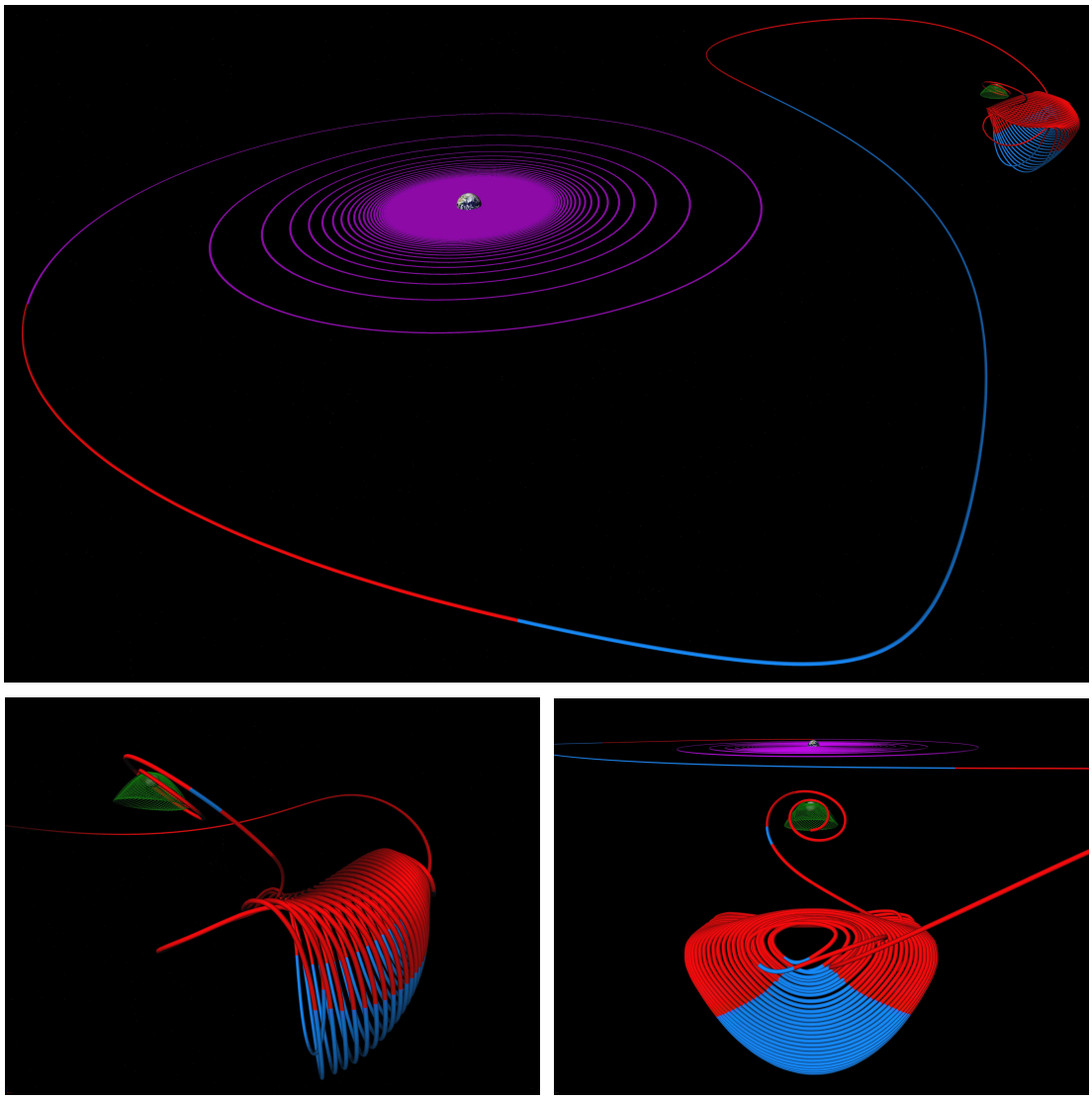


Figure 8. Feasible Three-Phase Trajectory.

The overall coverage time for the mission is limited by the amount of available propellant, and thus, the mass-time history plots in Figure 9 offer additional insight into the performance. In the top plot in Figure 9, the overall spacecraft mass for all three phases appears (with vertical dashed lines separating the phases). Recall that the initial guess for the coverage orbit assumes equal time spacing along all of the thrusting and coasting arcs, thus, the relatively piecewise linear decrease in mass over time is not surprising during phase #2. Although the coast times gradually increase to maintain continuity, the thrust times maintain a similar interval. In the middle plot in Figure 9, the mass performance for phase #1 is clear and, as expected, a significant translunar coast period is apparent during the transit between the two primaries. Finally, the bottom plot in Figure 9 reveals

that only a very short coast time is necessary to achieve insertion into a stable lunar frozen orbit. Since the feasible three-phase solution is used as an initial guess for optimization, Figure 9 also serves as a basis for comparison with the direct transcription procedure that manipulates the design variables to adjust the relative phase of the thrusting intervals.

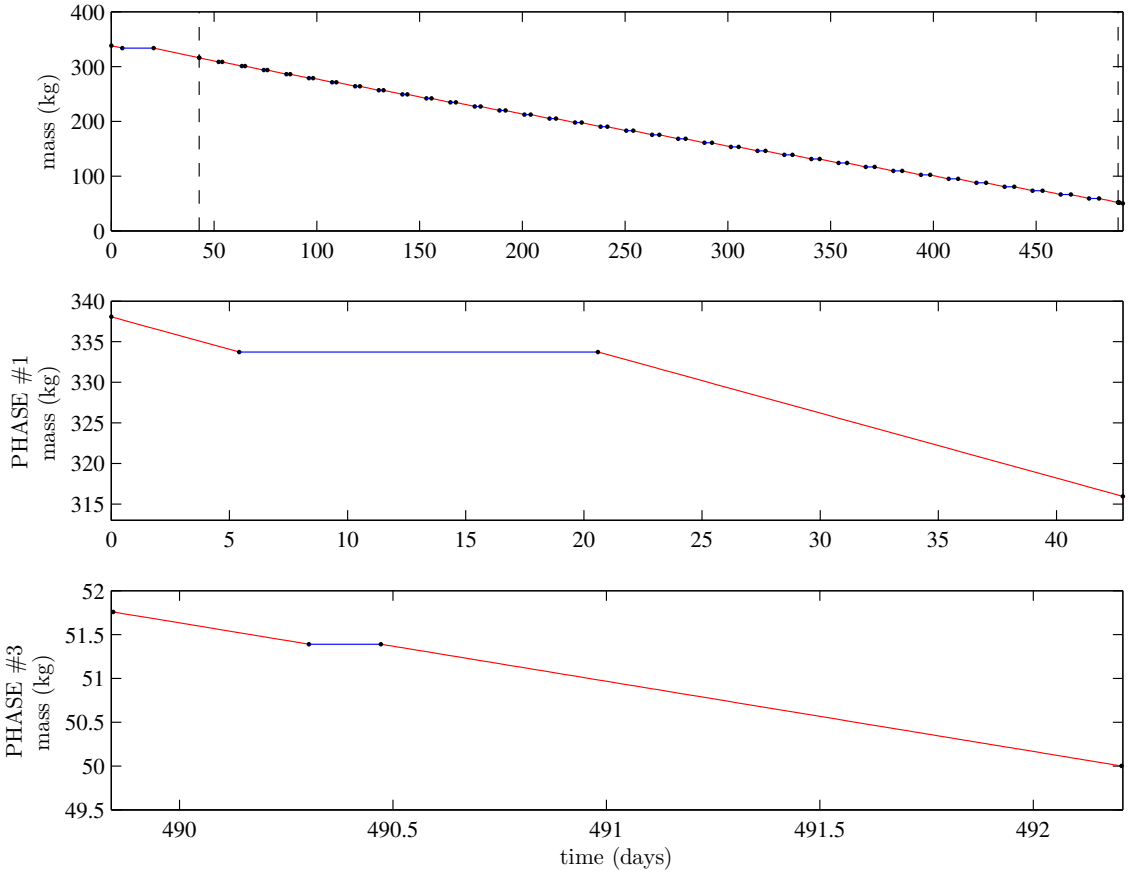


Figure 9. Spacecraft Mass for the Three-Phase Feasible Solution, Including Phase #1 and #3.

Optimal Solution. The direct transcription procedure is successfully implemented using the feasible three-phase solution as an initial condition. While the feasible solution is based on an existing trajectory that is near the initial guess, the direct transcription method iterates to produce a trajectory that minimizes the objective function in Eq. (19), or, in other words, maximizes the coverage time during phase #2. In total, an additional 107.14 days are added to the coverage orbit with this thrusting and coasting structure. The optimization procedure produces a significant variation from the initial guess. (See Figure 10.) The new trajectory, in comparison to the feasible solution, appears to lie closer to the minimum elevation angle boundary constraint. Such a result is not surprising considering, from Figure 4, that this region corresponds to lower required thrust acceleration magnitudes to maintain the pole-sitter position. Thus, when the thrust acceleration magnitude is small early in the trajectory, the most effective arcs are located in the regions closer to L_2. Although this behavior is somewhat intuitive in hindsight, it is remarkable that an initial guess that is relatively far from the optimal solution is automatically discovered by the method. The optimal solution inserts into the coverage orbit after a 236.30 day transfer, and achieves a total coverage orbit time of

554.18 days. Surprisingly, only an additional 8.90 kg of mass are required to achieve this orbit in comparison to the feasible solution. Considering that a mere 3% increase in fuel mass consumption yields a 24% longer coverage phase suggests that the optimal trajectory is primarily exploiting natural (uncontrolled) dynamics. In fact, optimization adds only a 3% increase in the total thrust time for phase #2, whereas there is a 82% increase in coast time.

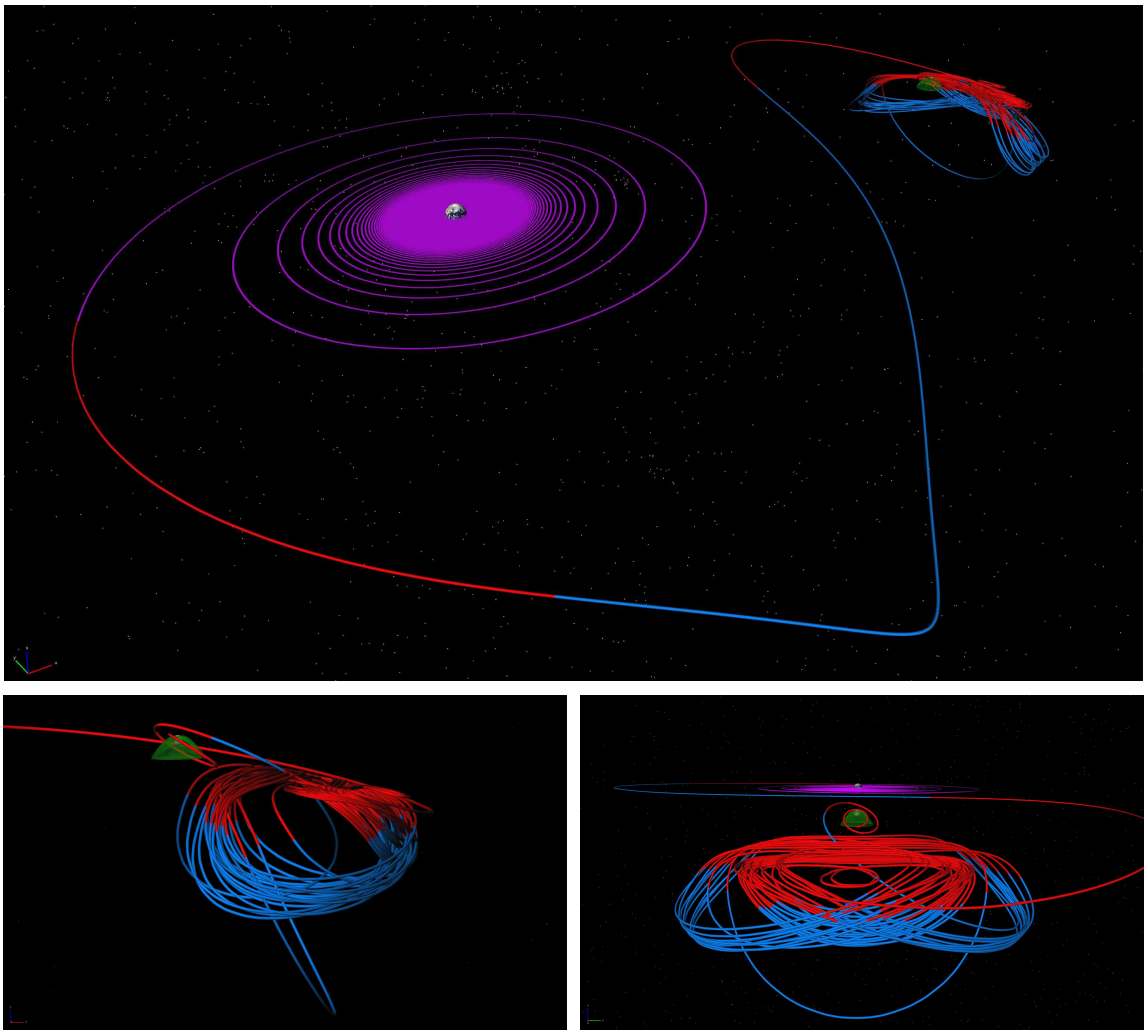


Figure 10. Optimal Three-Phase Trajectory.

Further insight into the behavior of the optimal solution is available Figure 11. Most obvious is that, compared to Figure 9, the thrust times for both phase #1 and phase #3 are shortened to maximize the total time in phase #2. In fact, the optimizer reduces the first thrust arc in phase #3 to zero days, which is the equivalent of shifting it into phase #2. In phase #2, the global trend is less linear compared to the feasible solution. For this phase, it is more effective to thrust over longer duration arcs early in the coverage orbit when the thrust acceleration magnitude is lower. Then, when the thrust acceleration magnitude is higher, near the end of the trajectory, less fuel is expended since it is only necessary to thrust in short bursts and coast during longer intervals. A useful comparison of the thrusting and coasting intervals between the feasible and optimal solutions appears in Figure 12.

The time for each optimal coast arc always lasts 2-5 days longer than the corresponding feasible coast arc. This amounts to a total increase in coverage time of 96.03 days, just due to coasting. (Compare this to the total increase in thrusting time of 11.11 days.) From Figure 12 it is clear that the optimal solution emphasizes the early long-thrusting arcs, and then, later, thrusting arcs are actually shorter than coasting intervals. In contrast, this exchange in thrust and coast time never occurs in the feasible solution. Clearly, this solution supports the hypothesis deduced by visualizing the approximate gravity field in Figure 4. Although the time increase along the optimal solution is highly desirable, the well-behaved thrusting and coasting time intervals associated with the feasible solution may offer advantages operationally for implementation. However, the optimal solution may possesses advantages as well, for example, even though the final thrust arc is only 0.34 days, most thrust arcs last longer than 9 days. In fact, the duration of the first thrust arc is over 30 days. For a summary and further numerical comparison between the two solutions, see Table 2.

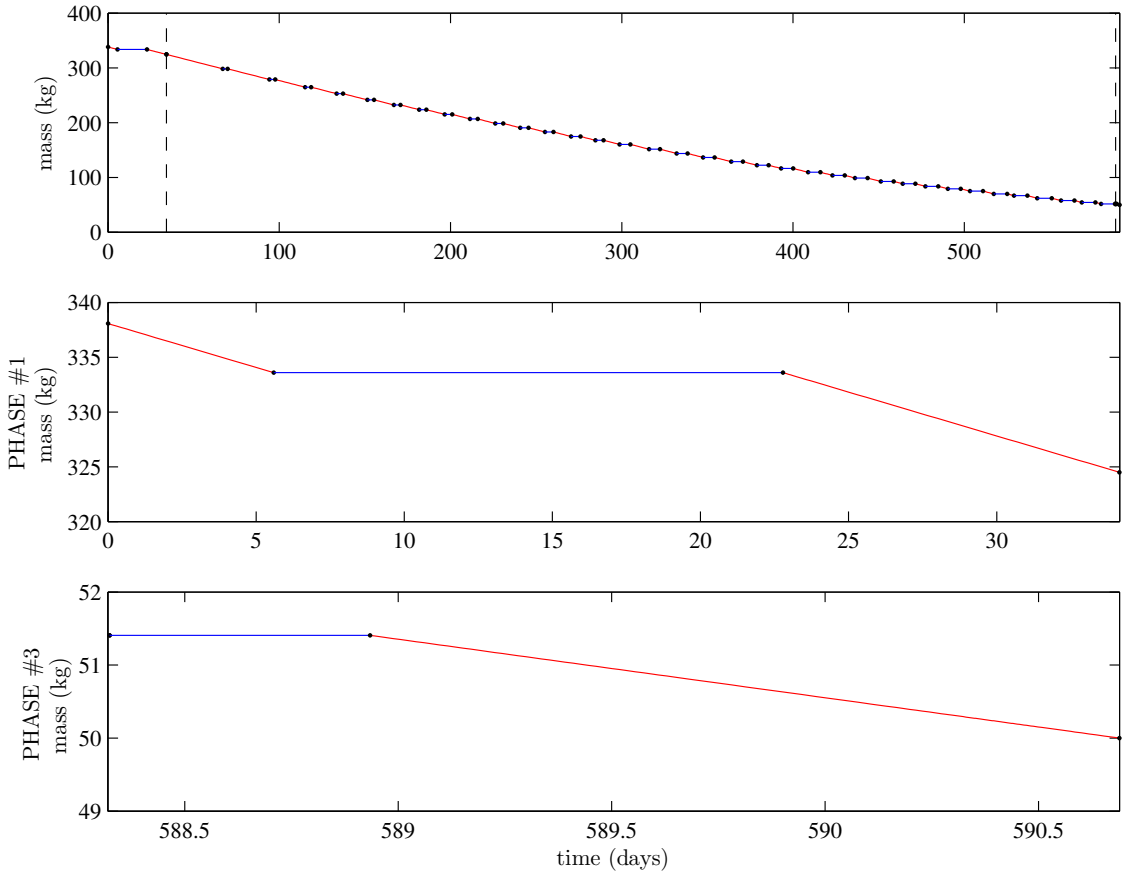


Figure 11. Spacecraft Mass for the Three-Phase Optimal Solution, Including Phase #1 and #3..

CONCLUSION

A general procedure involving collocation and direct transcription is successfully used to design trajectories for a low-thrust lunar pole-sitter mission. End-to-end trajectory design, in particular for the coverage orbit, is very challenging, and emphasis is placed on building the capability to *discover* new orbits. The method is also meant for easily combining several phases of flight, imple-

menting path constraints, exploring the available design space, and adding a wide range of objective functions for minimization. Implementation of this approach offers a natural means to explore and optimize feasible solutions. Although little intuition into either the feasible or optimal trajectories is initially available, investigating the magnitude of the three-body gravity gradient in the region of interest provides information regarding coverage limits and capabilities, as well as a nearly guess-free initial condition for a potential pole-sitter trajectory. The gravity potential information also supplies a means of interpreting the optimal solutions *a posteriori*. The resulting coverage orbits are nearly periodic solutions that appear to lie on a displaced family of L_1 and L_2 southern halo orbits. In general, solutions are easily extended beyond one year with the implementation of thrusting arcs and coasting arcs. This preliminary study demonstrates that low-thrust propulsion offers continuous lunar south pole coverage with a single spacecraft. The results are applicable to other systems, as well. Higher fidelity study of the lunar mission application, including ephemeris models, and a more accurate model for the engine thrust, is warranted for further validation of the concept.

Table 2. Performance Comparison of Feasible and Optimal Three-Phase Solutions.

		Feasible	Optimal
PHASE #1	No. of Thrust Arcs	3	3
	Fuel Mass Consumed (kg)	22.13	13.58
	Min. Thrust Arc Duration (days)	5.41	5.59
	Total Time (days)	42.81	34.14
PHASE #2	No. of Thrust Arcs	35	35
	Fuel Mass Consumed (kg)	264.19	273.09
	Avg. Thrust Arc Duration (days)	9.16	9.47
	Min. Thrust Arc Duration (days)	8.85	0.34
	Total Time (days)	447.04	554.18
	Min. Altitude (km)	32,400	24,750
	Max. Altitude (km)	80,000	100,000
	Min. Elevation Angle (°)	13.0	13.0
	Max. Elevation Angle (°)	79.5	60.5
PHASE #3	No. of Thrust Arcs	3	3
	Fuel Mass Consumed (kg)	1.76	1.41
	Min. Thrust Arc Duration (days)	0.46	0.00
	Total Time (days)	2.36	2.37

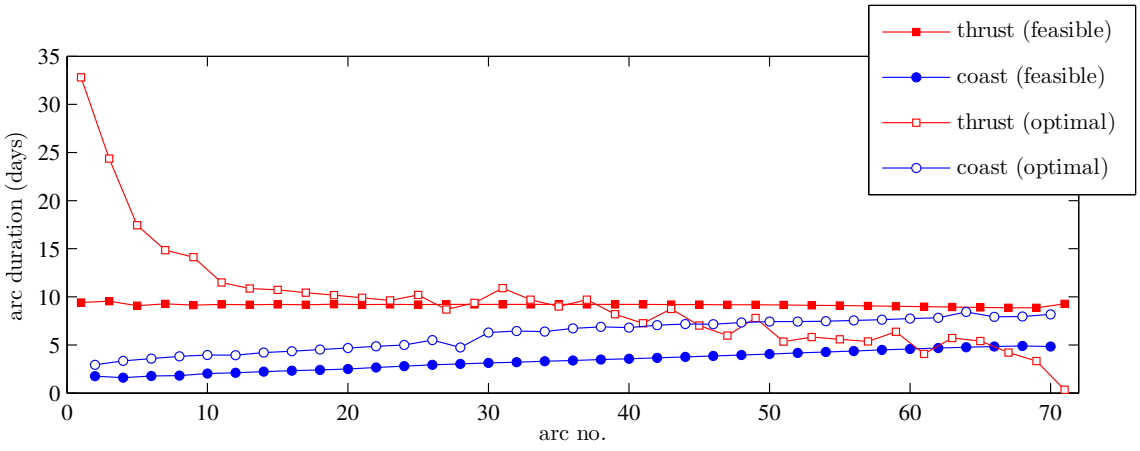


Figure 12. Arc Duration Comparison Between Feasible and Optimal Three-Phase Solutions.

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